

Chapter 4: Vector Spaces

All Worked Examples with Step-by-Step Solutions

Compiled for study. Equations and working lines are separated for readability.

Chapter 4: Vector Spaces

Step-by-step worked examples compiled from the Chapter 4 tutoring session based on Ron Larson, Elementary Linear Algebra, 8th ed.

This PDF focuses only on examples and their working steps. Formulas and working lines are separated for easier study.

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4.1 Vectors in $\mathbb{R}^n$

Example 4.1.1 - Representing vectors in the plane

Represent these vectors:

$$\begin{aligned}u &= (2, 3) \\v &= (-1, 2)\end{aligned}$$

Solution

A vector in the plane is drawn as an arrow starting at the origin and ending at the point named by its components.

For u :

$$u = (2, 3)$$

Start at:

$$(0, 0)$$

End at:

$$(2, 3)$$

So u means move 2 units right and 3 units up.

For v :

$$v = (-1, 2)$$

Start at:

$$(0, 0)$$

End at:

$$(-1, 2)$$

So v means move 1 unit left and 2 units up.

Final answers:

$$\begin{aligned}u &: \text{arrow from } (0, 0) \text{ to } (2, 3) \\v &: \text{arrow from } (0, 0) \text{ to } (-1, 2)\end{aligned}$$

Example 4.1.2 - Adding two vectors in the plane

Find each vector sum.

$$\begin{aligned}\text{(a) } u &= (1, 4), & v &= (2, -2) \\ \text{(b) } u &= (3, -2), & v &= (-3, 2) \\ \text{(c) } u &= (2, 1), & v &= (0, 0)\end{aligned}$$

Solution

Use componentwise addition:

$$(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$

Part (a):

$$\begin{aligned} u + v &= (1, 4) + (2, -2) \\ &= (1 + 2, 4 + (-2)) \\ &= (3, 2) \end{aligned}$$

Part (b):

$$\begin{aligned} u + v &= (3, -2) + (-3, 2) \\ &= (3 + (-3), -2 + 2) \\ &= (0, 0) \end{aligned}$$

The vector $(0, 0)$ is the zero vector.

Part (c):

$$\begin{aligned} u + v &= (2, 1) + (0, 0) \\ &= (2 + 0, 1 + 0) \\ &= (2, 1) \end{aligned}$$

Final answers:

- (a) $(3, 2)$
- (b) $(0, 0)$
- (c) $(2, 1)$

Example 4.1.3 - Operations with vectors in the plane

Let:

$$\begin{aligned} v &= (-2, 5) \\ u &= (3, 4) \end{aligned}$$

Find:

- (a) $(1/2)v$
- (b) $u - v$
- (c) $(1/2)v + u$

Solution

Part (a):

$$\begin{aligned} (1/2)v &= (1/2)(-2, 5) \\ &= ((1/2)(-2), (1/2)(5)) \\ &= (-1, 5/2) \end{aligned}$$

Part (b):

$$\begin{aligned} u - v &= (3, 4) - (-2, 5) \\ &= (3 - (-2), 4 - 5) \\ &= (3 + 2, -1) \\ &= (5, -1) \end{aligned}$$

Part (c):

Use the result from part (a):

$$(1/2)v = (-1, 5/2)$$

Then:

$$\begin{aligned}(1/2)v + u &= (-1, 5/2) + (3, 4) \\ &= (-1 + 3, 5/2 + 4) \\ &= (2, 5/2 + 8/2) \\ &= (2, 13/2)\end{aligned}$$

Final answers:

- (a) $(-1, 5/2)$
- (b) $(5, -1)$
- (c) $(2, 13/2)$

Example 4.1.4 - Vector operations in $\mathbb{R}^3$

Let:

$$\begin{aligned}u &= (-1, 0, 1) \\ v &= (2, -1, 5)\end{aligned}$$

Find:

- (a) $u + v$
- (b) $2u$
- (c) $v - 2u$

Solution

Part (a):

$$\begin{aligned}u + v &= (-1, 0, 1) + (2, -1, 5) \\ &= (-1 + 2, 0 + (-1), 1 + 5) \\ &= (1, -1, 6)\end{aligned}$$

Part (b):

$$\begin{aligned}2u &= 2(-1, 0, 1) \\ &= (2(-1), 2(0), 2(1)) \\ &= (-2, 0, 2)\end{aligned}$$

Part (c):

From part (b):

$$2u = (-2, 0, 2)$$

Then:

$$\begin{aligned}v - 2u &= (2, -1, 5) - (-2, 0, 2) \\ &= (2 - (-2), -1 - 0, 5 - 2) \\ &= (4, -1, 3)\end{aligned}$$

Final answers:

- (a) $(1, -1, 6)$
- (b) $(-2, 0, 2)$
- (c) $(4, -1, 3)$

Example 4.1.5 - Vector operations in $\mathbb{R}^4$

Let:

$$\begin{aligned}u &= (2, -1, 5, 0) \\v &= (4, 3, 1, -1) \\w &= (-6, 2, 0, 3)\end{aligned}$$

Find x from each equation:

$$\begin{aligned}\text{(a)} \quad x &= 2u - (v + 3w) \\ \text{(b)} \quad 3(x + w) &= 2u - v + x\end{aligned}$$

Solution

Part (a):

Start with:

$$x = 2u - (v + 3w)$$

Distribute the negative sign:

$$x = 2u - v - 3w$$

Compute each part:

$$\begin{aligned}2u &= 2(2, -1, 5, 0) \\ &= (4, -2, 10, 0)\end{aligned}$$

$$\begin{aligned}3w &= 3(-6, 2, 0, 3) \\ &= (-18, 6, 0, 9)\end{aligned}$$

Substitute:

$$x = (4, -2, 10, 0) - (4, 3, 1, -1) - (-18, 6, 0, 9)$$

Compute component by component:

$$\begin{aligned}\text{first component:} \quad & 4 - 4 - (-18) = 18 \\ \text{second component:} \quad & -2 - 3 - 6 = -11 \\ \text{third component:} \quad & 10 - 1 - 0 = 9 \\ \text{fourth component:} \quad & 0 - (-1) - 9 = -8\end{aligned}$$

So:

$$x = (18, -11, 9, -8)$$

Part (b):

Start with:

$$3(x + w) = 2u - v + x$$

Distribute 3:

$$3x + 3w = 2u - v + x$$

Move x terms to one side:

$$3x - x = 2u - v - 3w$$

So:

$$2x = 2u - v - 3w$$

Divide by 2:

$$x = (1/2)(2u - v - 3w)$$

From part (a):

$$2u - v - 3w = (18, -11, 9, -8)$$

Therefore:

$$\begin{aligned} x &= (1/2)(18, -11, 9, -8) \\ &= (9, -11/2, 9/2, -4) \end{aligned}$$

Final answers:

$$\begin{aligned} \text{(a) } x &= (18, -11, 9, -8) \\ \text{(b) } x &= (9, -11/2, 9/2, -4) \end{aligned}$$

Example 4.1.6 - Writing a vector as a linear combination

Let:

$$\begin{aligned} x &= (-1, -2, -2) \\ u &= (0, 1, 4) \\ v &= (-1, 1, 2) \\ w &= (3, 1, 2) \end{aligned}$$

Find scalars a, b, and c such that:

$$x = au + bv + cw$$

Solution

Write the equation:

$$(-1, -2, -2) = a(0, 1, 4) + b(-1, 1, 2) + c(3, 1, 2)$$

Multiply scalars into vectors:

$$\begin{aligned} a(0, 1, 4) &= (0, a, 4a) \\ b(-1, 1, 2) &= (-b, b, 2b) \\ c(3, 1, 2) &= (3c, c, 2c) \end{aligned}$$

Add the right side:

$$\begin{aligned} (0, a, 4a) + (-b, b, 2b) + (3c, c, 2c) \\ = (-b + 3c, a + b + c, 4a + 2b + 2c) \end{aligned}$$

Equate components:

$$\begin{aligned} -b + 3c &= -1 \\ a + b + c &= -2 \\ 4a + 2b + 2c &= -2 \end{aligned}$$

Solve the first equation for b:

$$-b + 3c = -1$$

$$\begin{aligned}-b &= -1 - 3c \\ b &= 1 + 3c\end{aligned}$$

Use the second equation:

$$\begin{aligned}a + b + c &= -2 \\ a + (1 + 3c) + c &= -2 \\ a + 1 + 4c &= -2 \\ a &= -3 - 4c\end{aligned}$$

Substitute into the third equation:

$$\begin{aligned}4(-3 - 4c) + 2(1 + 3c) + 2c &= -2 \\ -12 - 16c + 2 + 6c + 2c &= -2 \\ -10 - 8c &= -2 \\ -8c &= 8 \\ c &= -1\end{aligned}$$

Then:

$$\begin{aligned}b &= 1 + 3(-1) \\ &= -2\end{aligned}$$

and:

$$\begin{aligned}a &= -3 - 4(-1) \\ &= 1\end{aligned}$$

Final answer:

$$a = 1, b = -2, c = -1$$

So:

$$x = u - 2v - w$$

Check:

$$\begin{aligned}u - 2v - w &= (0, 1, 4) - 2(-1, 1, 2) - (3, 1, 2) \\ &= (0, 1, 4) + (2, -2, -4) - (3, 1, 2) \\ &= (2, -1, 0) - (3, 1, 2) \\ &= (-1, -2, -2)\end{aligned}$$

4.2 Vector Spaces

Example 4.2.1 - $\mathbb{R}^2$ is a vector space with standard operations

Show that $\mathbb{R}^2$ is a vector space under ordinary vector addition and scalar multiplication.

Solution

A general vector in $\mathbb{R}^2$ is:

$$u = (u_1, u_2)$$

Another vector is:

$$v = (v_1, v_2)$$

Addition is:

$$u + v = (u_1 + v_1, u_2 + v_2)$$

Because sums of real numbers are real, both components are real. Therefore:

$$u + v \text{ is in } \mathbb{R}^2$$

Scalar multiplication is:

$$\begin{aligned} cu &= c(u_1, u_2) \\ &= (cu_1, cu_2) \end{aligned}$$

Because products of real numbers are real:

$$cu \text{ is in } \mathbb{R}^2$$

The remaining vector-space axioms follow from ordinary real-number arithmetic.

Final conclusion:

$$\mathbb{R}^2 \text{ is a vector space.}$$

Example 4.2.2 - $\mathbb{R}^n$ is a vector space

Show that $\mathbb{R}^n$ is a vector space using standard operations.

Solution

Let:

$$\begin{aligned} u &= (u_1, u_2, \dots, u_n) \\ v &= (v_1, v_2, \dots, v_n) \end{aligned}$$

Addition is:

$$u + v = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

Every component is real, so the sum is in $\mathbb{R}^n$.

Scalar multiplication is:

$$\mathbf{c} = (c_1, c_2, \dots, c_n)$$

Every component is real, so $\mathbf{c}$ is in $\mathbb{R}^n$.

The other eight axioms follow from properties of real numbers.

Final conclusion:

$\mathbb{R}^n$ is a vector space.

Example 4.2.3 - The space of all 2 by 3 matrices

Show that the set $M_{\{2,3\}}$ of all 2 by 3 matrices is a vector space.

Solution

Let:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

Add component by component:

$$A + B = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \end{bmatrix}$$

This is still a 2 by 3 matrix.

Now multiply by a scalar c :

$$cA = \begin{bmatrix} ca_{11} & ca_{12} & ca_{13} \\ ca_{21} & ca_{22} & ca_{23} \end{bmatrix}$$

This is still a 2 by 3 matrix.

The zero vector is the zero matrix:

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The additive inverse is:

$$-A = \begin{bmatrix} -a_{11} & -a_{12} & -a_{13} \\ -a_{21} & -a_{22} & -a_{23} \end{bmatrix}$$

Final conclusion:

$M_{\{2,3\}}$ is a vector space.

Example 4.2.4 - The polynomial space P_2

Show that P_2 , the set of all polynomials of degree 2 or less, is a vector space.

Solution

A general polynomial in P_2 is:

$$p(x) = a_0 + a_1x + a_2x^2$$

Another polynomial is:

$$q(x) = b_0 + b_1x + b_2x^2$$

Add them:

$$\begin{aligned} p(x) + q(x) \\ = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 \end{aligned}$$

The result is still degree 2 or less.

Multiply by scalar c :

$$\begin{aligned} cp(x) &= c(a_0 + a_1x + a_2x^2) \\ &= ca_0 + ca_1x + ca_2x^2 \end{aligned}$$

The result is still degree 2 or less.

The zero vector is the zero polynomial:

$$0(x) = 0$$

The additive inverse is:

$$-p(x) = -a_0 - a_1x - a_2x^2$$

Final conclusion:

$$P_2 \text{ is a vector space.}$$

Example 4.2.5 - Continuous functions as a vector space

Show that $C[a,b]$, the set of continuous functions on $[a,b]$, is a vector space.

Solution

Let f and g be continuous on $[a,b]$.

Closure under addition:

$$(f + g)(x) = f(x) + g(x)$$

The sum of continuous functions is continuous, so $f + g$ is in $C[a,b]$.

Closure under scalar multiplication:

$$(cf)(x) = c f(x)$$

A scalar multiple of a continuous function is continuous, so cf is in $C[a,b]$.

The zero vector is the zero function:

$$0(x) = 0$$

The additive inverse is:

$$-f(x)$$

Final conclusion:

$C[a,b]$ is a vector space.

Example 4.2.6 - The integers are not a vector space over the real numbers

Show that $\mathbb{Z}$, the set of integers, is not a vector space over the real numbers.

Solution

Use standard addition and scalar multiplication.

Choose an integer:

$$1 \text{ is in } \mathbb{Z}$$

Choose a real scalar:

$$c = 1/2$$

Multiply:

$$(1/2)(1) = 1/2$$

But:

$$1/2 \text{ is not an integer}$$

So $\mathbb{Z}$ is not closed under scalar multiplication.

Final conclusion:

$$\mathbb{Z} \text{ is not a vector space over the real numbers.}$$

Example 4.2.7 - Polynomials of degree exactly 2 are not a vector space

Show that the set of all polynomials of degree exactly 2 is not a vector space.

Solution

Take two degree-2 polynomials:

$$\begin{aligned} p(x) &= x^2 \\ q(x) &= 1 + x - x^2 \end{aligned}$$

Add:

$$\begin{aligned} p(x) + q(x) &= x^2 + (1 + x - x^2) \\ &= 1 + x \end{aligned}$$

The result:

$$1 + x$$

has degree 1, not degree 2.

So the set is not closed under addition.

Also, scalar multiplication by 0 gives:

$$0 * x^2 = 0$$

The zero polynomial is not degree exactly 2.

Final conclusion:

The set of all polynomials of degree exactly 2 is not a vector space.

Example 4.2.8 - A set with unusual scalar multiplication is not a vector space

Let $V = \mathbb{R}^2$ with ordinary addition, but define scalar multiplication by:

$$c(x_1, x_2) = (cx_1, 0)$$

Show that V is not a vector space.

Solution

A vector space must satisfy:

$$1u = u$$

Choose:

$$u = (1, 1)$$

Using the given scalar multiplication:

$$\begin{aligned} 1(1, 1) &= (1*1, 0) \\ &= (1, 0) \end{aligned}$$

But:

$$(1, 0) \neq (1, 1)$$

So the scalar identity axiom fails.

Final conclusion:

V is not a vector space.

4.3 Subspaces of Vector Spaces

Example 4.3.1 - A subspace of $\mathbb{R}^3$

Show that:

$$W = \{ (x_1, 0, x_3) : x_1 \text{ and } x_3 \text{ are real} \}$$

is a subspace of $\mathbb{R}^3$.

Solution

Step 1: Check nonempty.

Choose:

$$x_1 = 0, x_3 = 0$$

Then:

$$(x_1, 0, x_3) = (0, 0, 0)$$

So W contains the zero vector.

Step 2: Check closure under addition.

Let:

$$u = (x_1, 0, x_3)$$

$$v = (y_1, 0, y_3)$$

Then:

$$\begin{aligned} u + v &= (x_1 + y_1, 0 + 0, x_3 + y_3) \\ &= (x_1 + y_1, 0, x_3 + y_3) \end{aligned}$$

The second component is still 0, so $u + v$ is in W .

Step 3: Check scalar multiplication.

Let c be any scalar:

$$\begin{aligned} cu &= c(x_1, 0, x_3) \\ &= (cx_1, 0, cx_3) \end{aligned}$$

The second component is still 0, so cu is in W .

Final conclusion:

$$W \text{ is a subspace of } \mathbb{R}^3.$$

Example 4.3.2 - Symmetric matrices form a subspace

Show that the set W of all 2 by 2 symmetric matrices is a subspace of $M_{\{2,2\}}$.

Solution

A symmetric matrix satisfies:

$$A^T = A$$

A general 2 by 2 symmetric matrix is:

$$A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

Step 1: Zero matrix.

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Since $O^T = O$, the zero matrix is symmetric.

Step 2: Addition.

Let A and B be symmetric:

$$\begin{aligned} A^T &= A \\ B^T &= B \end{aligned}$$

Then:

$$\begin{aligned} (A + B)^T &= A^T + B^T \\ &= A + B \end{aligned}$$

So $A + B$ is symmetric.

Step 3: Scalar multiplication.

Let c be a scalar:

$$\begin{aligned} (cA)^T &= cA^T \\ &= cA \end{aligned}$$

So cA is symmetric.

Final conclusion:

The set of 2 by 2 symmetric matrices is a subspace of $M_{\{2,2\}}$.

Example 4.3.3 - Singular matrices do not form a subspace

Show that the set of singular 2 by 2 matrices is not a subspace of $M_{\{2,2\}}$.

Solution

Use two singular matrices:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Check A:

$$\det(A) = 1(0) - 0(0) = 0$$

So A is singular.

Check B:

$$\det(B) = 0(1) - 0(0) = 0$$

So B is singular.

Add:

$$\begin{aligned} A + B &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

This is the identity matrix.

$$\det(I) = 1$$

So A + B is not singular.

Final conclusion:

The set of singular 2 by 2 matrices is not closed under addition.
Therefore it is not a subspace.

Example 4.3.4 - The first quadrant is not a subspace

Show that:

$$W = \{ (x_1, x_2) : x_1 \geq 0 \text{ and } x_2 \geq 0 \}$$

is not a subspace of $\mathbb{R}^2$.

Solution

The zero vector is in W:

$$(0, 0) \text{ is in } W$$

The set is closed under addition because the sum of nonnegative numbers is nonnegative.

But scalar multiplication fails.

Choose:

$$u = (1, 1)$$

Then u is in W.

Choose scalar:

$$c = -1$$

Then:

$$\begin{aligned} cu &= -1(1, 1) \\ &= (-1, -1) \end{aligned}$$

But (-1, -1) is not in W.

Final conclusion:

W is not a subspace of $\mathbb{R}^2$.

Example 4.3.5 - Nested function subspaces

Let:

W1 = polynomial functions on $[0,1]$
W2 = differentiable functions on $[0,1]$
W3 = continuous functions on $[0,1]$
W4 = integrable functions on $[0,1]$
W5 = all functions on $[0,1]$

Explain why:

W1 subset W2 subset W3 subset W4 subset W5

Solution

A polynomial function is differentiable.

Every differentiable function is continuous.

Every continuous function on a closed interval is integrable.

Every integrable function is a function.

So:

W1 subset W2 subset W3 subset W4 subset W5

Each smaller space is a subspace of the larger spaces because it is closed under addition and scalar multiplication.

For example, if p and q are polynomials, then:

p + q is a polynomial
cp is a polynomial

Final conclusion:

The spaces are nested subspaces.

Example 4.3.6 - Lines in $\mathbb{R}^2$

Determine whether each line is a subspace of $\mathbb{R}^2$:

- (a) $x + 2y = 0$
(b) $x + 2y = 1$

Solution

Part (a):

Solve:

$$\begin{aligned}x + 2y &= 0 \\x &= -2y\end{aligned}$$

Let:

$$y = t$$

Then:

$$x = -2t$$

So every point has the form:

$$(x, y) = (-2t, t) = t(-2, 1)$$

This is a line through the origin, so it is closed under addition and scalar multiplication.

Conclusion:

$$x + 2y = 0 \text{ is a subspace of } \mathbb{R}^2.$$

Part (b):

Check the zero vector:

$$0 + 2(0) = 0$$

But the equation requires:

$$x + 2y = 1$$

Since:

$$0 \neq 1$$

$(0,0)$ is not on the line.

Conclusion:

$$x + 2y = 1 \text{ is not a subspace of } \mathbb{R}^2.$$

Example 4.3.7 - The unit circle is not a subspace

Show that:

$$x^2 + y^2 = 1$$

is not a subspace of $\mathbb{R}^2$.

Solution

Choose two points on the circle:

$$\begin{aligned} (1, 0) \\ (0, 1) \end{aligned}$$

Check:

$$\begin{aligned} 1^2 + 0^2 &= 1 \\ 0^2 + 1^2 &= 1 \end{aligned}$$

Both are in the set.

Add them:

$$(1, 0) + (0, 1) = (1, 1)$$

Check whether (1,1) is on the circle:

$$1^2 + 1^2 = 2$$

But the circle requires:

$$x^2 + y^2 = 1$$

So (1,1) is not in the set.

Final conclusion:

The unit circle is not closed under addition.
Therefore it is not a subspace.

Example 4.3.8 - Determining subspaces of $\mathbb{R}^3$

Determine whether each set is a subspace of $\mathbb{R}^3$:

(a) $W = \{ (x_1, x_2, 1) \}$

(b) $W = \{ (x_1, x_1 + x_3, x_3) \}$

Solution

Part (a):

Every vector has third component equal to 1.

The zero vector is:

$$(0, 0, 0)$$

Its third component is 0, not 1.

So:

$$(0, 0, 0) \text{ is not in } W$$

Conclusion:

W is not a subspace of $\mathbb{R}^3$.

Part (b):

A vector in W has the form:

$$(x_1, x_1 + x_3, x_3)$$

Check zero vector:

$$x_1 = 0, x_3 = 0$$

Then:

$$(0, 0 + 0, 0) = (0, 0, 0)$$

So W is nonempty.

Check addition:

$$u = (a, a + b, b)$$

$$v = (c, c + d, d)$$

Then:

$$\begin{aligned} u + v &= (a + c, a + b + c + d, b + d) \\ &= (a + c, (a + c) + (b + d), b + d) \end{aligned}$$

This has the same form.

Check scalar multiplication:

$$\begin{aligned} ku &= k(a, a + b, b) \\ &= (ka, ka + kb, kb) \\ &= (ka, ka + kb, kb) \end{aligned}$$

This also has the same form.

Conclusion:

W is a subspace of $\mathbb{R}^3$.

4.4 Spanning Sets and Linear Independence

Example 4.4.1 - Examples of linear combinations

Part (a): Vectors in $\mathbb{R}^3$

Let:

$$\begin{aligned}v_1 &= (1, 3, 1) \\v_2 &= (0, 1, 2) \\v_3 &= (1, 0, -5)\end{aligned}$$

Show that v_1 is a linear combination of v_2 and v_3 .

Solution

Compute:

$$3v_2 + v_3 = 3(0, 1, 2) + (1, 0, -5)$$

Multiply:

$$3(0, 1, 2) = (0, 3, 6)$$

Add:

$$(0, 3, 6) + (1, 0, -5) = (1, 3, 1)$$

Since:

$$(1, 3, 1) = v_1$$

Final answer:

$$v_1 = 3v_2 + v_3$$

Part (b): Matrices

Let:

$$v_1 = \begin{bmatrix} 0 & 2 \\ 8 & 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -1 & 1 \\ 3 & 2 \end{bmatrix}$$

$$v_4 = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$$

Show:

$$v_1 = v_2 + 2v_3 - v_4$$

Compute:

$$2v_3 = \begin{bmatrix} -2 & 2 \\ 6 & 4 \end{bmatrix}$$

$$-v_4 = \begin{bmatrix} 2 & -1 \\ 0 & -3 \end{bmatrix}$$

Then:

$$\begin{aligned} v_2 + 2v_3 - v_4 &= \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} -2 & 2 \\ 6 & 4 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ 0 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 0 - 2 + 2 & 1 + 2 - 1 \\ 2 + 6 + 0 & 0 + 4 - 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 8 & 1 \end{bmatrix} \end{aligned}$$

Final answer:

$$v_1 = v_2 + 2v_3 - v_4$$

Example 4.4.2 - Finding a linear combination

Write:

$$w = (1, 1, 1)$$

as a linear combination of:

$$\begin{aligned} v_1 &= (1, 2, 3) \\ v_2 &= (0, 1, 2) \\ v_3 &= (-1, 0, 1) \end{aligned}$$

Solution

Set up:

$$(1, 1, 1) = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-1, 0, 1)$$

The right side is:

$$(c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3)$$

Equate components:

$$\begin{aligned} c_1 - c_3 &= 1 \\ 2c_1 + c_2 &= 1 \\ 3c_1 + 2c_2 + c_3 &= 1 \end{aligned}$$

Augmented matrix:

$$\begin{bmatrix} 1 & 0 & -1 & | & 1 \\ 2 & 1 & 0 & | & 1 \\ 3 & 2 & 1 & | & 1 \end{bmatrix}$$

Row reduction:

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1 \\ R_3 &\leftarrow R_3 - 3R_1 \\ &\begin{bmatrix} 1 & 0 & -1 & | & 1 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & -1 \\ 0 & 2 & 4 & | & -2 \end{bmatrix}$$

$$R3 \leftarrow R3 - 2R2$$

$$\begin{bmatrix} 1 & 0 & -1 & | & 1 \\ 0 & 1 & 2 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

So:

$$\begin{aligned} c1 - c3 &= 1 \\ c2 + 2c3 &= -1 \end{aligned}$$

Let:

$$c3 = t$$

Then:

$$\begin{aligned} c1 &= 1 + t \\ c2 &= -1 - 2t \end{aligned}$$

Choose $t = 1$:

$$\begin{aligned} c1 &= 2 \\ c2 &= -3 \\ c3 &= 1 \end{aligned}$$

Final answer:

$$w = 2v1 - 3v2 + v3$$

Check:

$$\begin{aligned} &2(1,2,3) - 3(0,1,2) + (-1,0,1) \\ &= (2,4,6) + (0,-3,-6) + (-1,0,1) \\ &= (1,1,1) \end{aligned}$$

Example 4.4.3 - A vector that cannot be written as a linear combination

Using:

$$\begin{aligned} v1 &= (1, 2, 3) \\ v2 &= (0, 1, 2) \\ v3 &= (-1, 0, 1) \end{aligned}$$

Determine whether:

$$w = (1, -2, 2)$$

can be written as a linear combination of $v1, v2, v3$.

Solution

Set up:

$$(1, -2, 2) = c1v1 + c2v2 + c3v3$$

This gives:

$$\begin{aligned}c_1 - c_3 &= 1 \\2c_1 + c_2 &= -2 \\3c_1 + 2c_2 + c_3 &= 2\end{aligned}$$

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array} \right]$$

Row reduce:

$$\begin{aligned}R_2 &\leftarrow R_2 - 2R_1 \\R_3 &\leftarrow R_3 - 3R_1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 2 & 4 & -1 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 7 \end{array} \right]$$

The last row means:

$$0 = 7$$

This is impossible.

Final conclusion:

w cannot be written as a linear combination of v_1, v_2, v_3 .

Example 4.4.4 - Standard spanning sets

Part (a): $\mathbb{R}^3$

Show that:

$$S = \{ (1,0,0), (0,1,0), (0,0,1) \}$$

spans $\mathbb{R}^3$.

Solution

Take any vector:

$$u = (u_1, u_2, u_3)$$

Write:

$$u = u_1(1,0,0) + u_2(0,1,0) + u_3(0,0,1)$$

Therefore every vector in $\mathbb{R}^3$ is a linear combination of vectors in S .

Conclusion:

$$S \text{ spans } \mathbb{R}^3.$$

Part (b): P₂

Show that:

$$S = \{1, x, x^2\}$$

spans P₂.

A general polynomial in P₂ is:

$$p(x) = a + bx + cx^2$$

Write:

$$p(x) = a(1) + b(x) + c(x^2)$$

Conclusion:

$$\{1, x, x^2\} \text{ spans } P_2.$$

Example 4.4.5 - A nonstandard spanning set of R³

Show that:

$$S = \{ (1,2,3), (0,1,2), (-2,0,1) \}$$

spans R³.

Solution

Let:

$$u = (u_1, u_2, u_3)$$

We need:

$$(u_1, u_2, u_3) = c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1)$$

This gives:

$$\begin{aligned} c_1 - 2c_3 &= u_1 \\ 2c_1 + c_2 &= u_2 \\ 3c_1 + 2c_2 + c_3 &= u_3 \end{aligned}$$

The coefficient matrix is:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute determinant:

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} - 0 \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} + (-2) \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \\ &= 1(1*1 - 0*2) + (-2)(2*2 - 1*3) \\ &= 1(1) + (-2)(1) \end{aligned}$$

$$= -1$$

Since:

$$\det(A) \neq 0$$

the system has a unique solution for every u in $\mathbb{R}^3$.

Final conclusion:

S spans $\mathbb{R}^3$.

Example 4.4.6 - A set that does not span $\mathbb{R}^3$

Show that:

$$S = \{ (1,2,3), (0,1,2), (-1,0,1) \}$$

does not span $\mathbb{R}^3$.

Solution

From Example 4.4.3, the vector:

$$w = (1, -2, 2)$$

cannot be written as a linear combination of the vectors in S .

If even one vector in $\mathbb{R}^3$ cannot be built from S , then S does not span $\mathbb{R}^3$.

Final conclusion:

S does not span $\mathbb{R}^3$.

Example 4.4.7 - Linearly dependent sets

Part (a)

$$S = \{ (1,2), (2,4) \}$$

Since:

$$(2,4) = 2(1,2)$$

the vectors are dependent.

A nontrivial dependence equation is:

$$-2(1,2) + (2,4) = (0,0)$$

Conclusion:

S is linearly dependent.

Part (b)

$$S = \{ (1,0), (0,1), (-2,5) \}$$

Compute:

$$\begin{aligned} & 2(1,0) - 5(0,1) + (-2,5) \\ &= (2,0) + (0,-5) + (-2,5) \\ &= (0,0) \end{aligned}$$

The coefficients are not all zero.

Conclusion:

S is linearly dependent.

Part (c)

$$S = \{ (0,0), (1,2) \}$$

Any set containing the zero vector is linearly dependent because:

$$1(0,0) + 0(1,2) = (0,0)$$

Conclusion:

S is linearly dependent.

Example 4.4.8 - Testing linear independence in $\mathbb{R}^3$

Determine whether:

$$S = \{ (1,2,3), (0,1,2), (-2,0,1) \}$$

is linearly independent.

Solution

Solve:

$$c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (0,0,0)$$

This gives:

$$\begin{aligned} c_1 - 2c_3 &= 0 \\ 2c_1 + c_2 &= 0 \\ 3c_1 + 2c_2 + c_3 &= 0 \end{aligned}$$

Augmented matrix:

$$\begin{bmatrix} 1 & 0 & -2 & | & 0 \\ 2 & 1 & 0 & | & 0 \\ 3 & 2 & 1 & | & 0 \end{bmatrix}$$

Row reduce:

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1 \\ R_3 &\leftarrow R_3 - 3R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & -2 & | & 0 \\ 0 & 1 & 4 & | & 0 \\ 0 & 2 & 7 & | & 0 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -2 & | & 0 \\ 0 & 1 & 4 & | & 0 \\ 0 & 0 & -1 & | & 0 \end{bmatrix}$$

From the last row:

$$c_3 = 0$$

Then:

$$c_2 = 0$$

$$c_1 = 0$$

Only the trivial solution exists.

Final conclusion:

S is linearly independent.

Example 4.4.9 - Testing linear independence in P_2

Determine whether:

$$S = \{ 1 + x - 2x^2, \quad 2 + 5x - x^2, \quad x + x^2 \}$$

is linearly independent.

Solution

Let:

$$v_1 = 1 + x - 2x^2$$

$$v_2 = 2 + 5x - x^2$$

$$v_3 = x + x^2$$

Solve:

$$c_1v_1 + c_2v_2 + c_3v_3 = 0$$

Substitute:

$$c_1(1 + x - 2x^2) + c_2(2 + 5x - x^2) + c_3(x + x^2) = 0$$

Collect coefficients:

$$\text{constant: } c_1 + 2c_2$$

$$x: \quad c_1 + 5c_2 + c_3$$

$$x^2: \quad -2c_1 - c_2 + c_3$$

Set each coefficient equal to zero:

$$c_1 + 2c_2 = 0$$

$$c_1 + 5c_2 + c_3 = 0$$

$$-2c_1 - c_2 + c_3 = 0$$

Augmented matrix:

$$\begin{bmatrix} 1 & 2 & 0 & | & 0 \\ 1 & 5 & 1 & | & 0 \\ -2 & -1 & 1 & | & 0 \end{bmatrix}$$

Row reduce:

$$\begin{aligned} R2 &\leftarrow R2 - R1 \\ R3 &\leftarrow R3 + 2R1 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 0 & | & 0 \\ 0 & 3 & 1 & | & 0 \\ 0 & 3 & 1 & | & 0 \end{bmatrix}$$

$$R3 \leftarrow R3 - R2$$

$$\begin{bmatrix} 1 & 2 & 0 & | & 0 \\ 0 & 3 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

There is a free variable.

From row 2:

$$3c_2 + c_3 = 0$$

Let:

$$c_2 = -t$$

Then:

$$c_3 = 3t$$

From row 1:

$$\begin{aligned} c_1 + 2c_2 &= 0 \\ c_1 = -2c_2 &= 2t \end{aligned}$$

Choose $t = 1$:

$$c_1 = 2, \quad c_2 = -1, \quad c_3 = 3$$

So:

$$2v_1 - v_2 + 3v_3 = 0$$

Final conclusion:

S is linearly dependent.

Example 4.4.10 - Testing linear independence in $M_{\{2,2\}}$

Determine whether the set is linearly independent:

$$v_1 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Solution

Solve:

$$c_1v_1 + c_2v_2 + c_3v_3 = 0$$

Compute:

$$c_1v_1 = \begin{bmatrix} 2c_1 & 0 \\ c_1 & c_1 \end{bmatrix}$$

$$c_2v_2 = \begin{bmatrix} 3c_2 & 2c_2 \\ 0 & c_2 \end{bmatrix}$$

$$c_3v_3 = \begin{bmatrix} c_3 & 2c_3 \\ 0 & 0 \end{bmatrix}$$

Add:

$$\begin{bmatrix} 2c_1 + 3c_2 + c_3 & 2c_2 + 2c_3 \\ c_1 & c_1 + c_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Equate entries:

$$\begin{aligned} 2c_1 + 3c_2 + c_3 &= 0 \\ 2c_2 + 2c_3 &= 0 \\ c_1 &= 0 \\ c_1 + c_2 &= 0 \end{aligned}$$

From $c_1 = 0$:

$$c_2 = 0$$

Then:

$$\begin{aligned} 2c_2 + 2c_3 &= 0 \\ 2(0) + 2c_3 &= 0 \\ c_3 &= 0 \end{aligned}$$

Only the trivial solution exists.

Final conclusion:

The set is linearly independent.

Example 4.4.11 - Testing linear independence in $M_{\{4,1\}}$

Determine whether:

$$\begin{aligned} v_1 &= \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} & v_2 &= \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} & v_3 &= \begin{bmatrix} 0 \\ 3 \\ 1 \\ -2 \end{bmatrix} & v_4 &= \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix} \end{aligned}$$

is linearly independent.

Solution

Set:

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$$

Equating components gives:

$$\begin{aligned} c_1 + c_2 &= 0 \\ c_2 + 3c_3 + c_4 &= 0 \\ -c_1 + c_3 - c_4 &= 0 \\ 2c_2 - 2c_3 + 2c_4 &= 0 \end{aligned}$$

The corresponding augmented matrix row-reduces to the identity matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix}$$

Therefore:

$$c_1 = c_2 = c_3 = c_4 = 0$$

Final conclusion:

The set is linearly independent.

Example 4.4.12 - Expressing one vector as a combination of others

From Example 4.4.9:

$$2v_1 - v_2 + 3v_3 = 0$$

Solve for v_2 .

Solution

Start:

$$2v_1 - v_2 + 3v_3 = 0$$

Move $-v_2$ to the other side:

$$2v_1 + 3v_3 = v_2$$

So:

$$v_2 = 2v_1 + 3v_3$$

Using the polynomials:

$$\begin{aligned} v_1 &= 1 + x - 2x^2 \\ v_3 &= x + x^2 \end{aligned}$$

Compute:

$$\begin{aligned} &2v_1 + 3v_3 \\ &= 2(1 + x - 2x^2) + 3(x + x^2) \\ &= 2 + 2x - 4x^2 + 3x + 3x^2 \\ &= 2 + 5x - x^2 \\ &= v_2 \end{aligned}$$

Final answer:

$$v_2 = 2v_1 + 3v_3$$

Example 4.4.13 - Two-vector shortcut

Determine whether each set is linearly independent.

(a) $\{ (1, 2, 0), (-2, 2, 1) \}$

(b) $\{ (4, -4, -2), (-2, 2, 1) \}$

Solution

For two vectors, the set is dependent if one vector is a scalar multiple of the other.

Part (a):

Suppose:

$$(-2, 2, 1) = c(1, 2, 0)$$

Then the third component would require:

$$1 = c \cdot 0 = 0$$

Impossible.

Conclusion:

The set is linearly independent.

Part (b):

Check:

$$-2(-2, 2, 1) = (4, -4, -2)$$

So one vector is a scalar multiple of the other.

Conclusion:

The set is linearly dependent.

4.5 Basis and Dimension

Example 4.5.1 - Standard basis for $\mathbb{R}^3$

Show that:

$$S = \{ (1,0,0), (0,1,0), (0,0,1) \}$$

is a basis for $\mathbb{R}^3$.

Solution

A basis must span and be linearly independent.

Spanning:

Let:

$$u = (u_1, u_2, u_3)$$

Then:

$$u = u_1(1,0,0) + u_2(0,1,0) + u_3(0,0,1)$$

So S spans $\mathbb{R}^3$.

Independence:

Solve:

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (0,0,0)$$

This gives:

$$(c_1, c_2, c_3) = (0,0,0)$$

So:

$$c_1 = c_2 = c_3 = 0$$

Only the trivial solution exists.

Final conclusion:

$$S \text{ is a basis for } \mathbb{R}^3.$$

Example 4.5.2 - A nonstandard basis for $\mathbb{R}^2$

Show that:

$$S = \{ (1,1), (1,-1) \}$$

is a basis for $\mathbb{R}^2$.

Solution

Let:

$$x = (x_1, x_2)$$

Try to write:

$$(x_1, x_2) = c_1(1, 1) + c_2(1, -1)$$

Right side:

$$\begin{aligned} c_1(1, 1) + c_2(1, -1) \\ = (c_1 + c_2, c_1 - c_2) \end{aligned}$$

Equate components:

$$\begin{aligned} c_1 + c_2 &= x_1 \\ c_1 - c_2 &= x_2 \end{aligned}$$

Add equations:

$$\begin{aligned} 2c_1 &= x_1 + x_2 \\ c_1 &= (x_1 + x_2)/2 \end{aligned}$$

Subtract equations:

$$\begin{aligned} 2c_2 &= x_1 - x_2 \\ c_2 &= (x_1 - x_2)/2 \end{aligned}$$

So every vector in $\mathbb{R}^2$ can be written as a linear combination of S .

Now check independence:

$$c_1(1, 1) + c_2(1, -1) = (0, 0)$$

This gives:

$$\begin{aligned} c_1 + c_2 &= 0 \\ c_1 - c_2 &= 0 \end{aligned}$$

Add:

$$\begin{aligned} 2c_1 &= 0 \\ c_1 &= 0 \end{aligned}$$

Then:

$$c_2 = 0$$

Final conclusion:

S is a basis for $\mathbb{R}^2$.

Example 4.5.3 - A nonstandard basis for $\mathbb{R}^3$

Show that:

$$S = \{ (1, 2, 3), (0, 1, 2), (-2, 0, 1) \}$$

is a basis for $\mathbb{R}^3$.

Solution

Put the vectors as columns:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute determinant:

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} + (-2) \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \\ &= 1(1 \cdot 1 - 0 \cdot 2) + (-2)(2 \cdot 2 - 1 \cdot 3) \\ &= 1 - 2 \\ &= -1 \end{aligned}$$

Since:

$$\det(A) \neq 0$$

the columns are linearly independent and span $\mathbb{R}^3$.

Final conclusion:

S is a basis for $\mathbb{R}^3$.

Example 4.5.4 - Standard basis for P_3

Show that:

$$S = \{1, x, x^2, x^3\}$$

is a basis for P_3 .

Solution

A general polynomial in P_3 is:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

Write:

$$p(x) = a_0(1) + a_1(x) + a_2(x^2) + a_3(x^3)$$

So S spans P_3 .

Check independence:

$$a_0(1) + a_1x + a_2x^2 + a_3x^3 = 0$$

The zero polynomial requires all coefficients to be zero:

$$a_0 = a_1 = a_2 = a_3 = 0$$

Final conclusion:

S is a basis for P_3 .

Example 4.5.5 - Standard basis for $M_{\{2,2\}}$

Show that the following set is a basis for $M_{\{2,2\}}$:

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Solution

A general 2 by 2 matrix is:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Write it as:

$$A = aE_{11} + bE_{12} + cE_{21} + dE_{22}$$

That expands to:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

So the set spans $M_{\{2,2\}}$.

For independence, solve:

$$c_1E_{11} + c_2E_{12} + c_3E_{21} + c_4E_{22} = 0$$

This gives:

$$\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

So:

$$c_1 = c_2 = c_3 = c_4 = 0$$

Final conclusion:

$$\text{The set is a basis for } M_{\{2,2\}}.$$

Example 4.5.6 - Uniqueness of basis representation

Let:

$$B = \{ (1,2,3), (0,1,2), (-2,0,1) \}$$

Explain why every vector in $\mathbb{R}^3$ has exactly one representation in this basis.

Solution

From Example 4.5.3, B is a basis.

That means:

$$\begin{aligned} B &\text{ spans } \mathbb{R}^3 \\ B &\text{ is linearly independent} \end{aligned}$$

Spanning guarantees that every vector can be written using B.

Linear independence guarantees that the representation is unique.

For a vector:

$$x = (x_1, x_2, x_3)$$

the equation:

$$x = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-2, 0, 1)$$

has coefficient matrix:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Because:

$$\det(A) = -1 \neq 0$$

the system has exactly one solution.

Final conclusion:

Every vector in $\mathbb{R}^3$ has exactly one coordinate representation relative to B .

Example 4.5.7 - Too many vectors cause dependence

Determine whether each set is linearly dependent.

- (a) S has 4 vectors in $\mathbb{R}^3$
- (b) S has 5 polynomials in P_3

Solution

Part (a):

$$\dim(\mathbb{R}^3) = 3$$

Any set with more than 3 vectors in $\mathbb{R}^3$ is linearly dependent.

Since S has 4 vectors:

S is linearly dependent.

Part (b):

$$\dim(P_3) = 4$$

Any set with more than 4 vectors in P_3 is linearly dependent.

Since S has 5 polynomials:

S is linearly dependent.

Example 4.5.8 - Explaining why a set is not a basis

Part (a)

Can:

$$S = \{ (3, 2, 1), (7, -1, 4) \}$$

be a basis for $\mathbb{R}^3$?

Solution

$\mathbb{R}^3$ has dimension:

$$\dim(\mathbb{R}^3) = 3$$

A basis for $\mathbb{R}^3$ must have exactly 3 vectors.

This set has only 2 vectors.

Conclusion:

S is not a basis for $\mathbb{R}^3$.

Part (b)

Can a set of 5 polynomials be a basis for P_3 ?

Solution

P_3 has dimension:

$$\dim(P_3) = 4$$

A basis for P_3 must have exactly 4 vectors.

A set of 5 vectors in P_3 is automatically dependent.

Conclusion:

A set of 5 polynomials cannot be a basis for P_3 .

Example 4.5.9 - Dimensions of subspaces of $\mathbb{R}^3$

Find the dimension of each subspace.

- (a) $W = \{ (d, c-d, c) : c, d \text{ real} \}$
- (b) $W = \{ (2b, b, 0) : b \text{ real} \}$

Solution

Part (a):

Start with:

$$(d, c-d, c)$$

Separate c-parts and d-parts:

$$(d, c-d, c) = (0, c, c) + (d, -d, 0)$$

Factor:

$$\begin{aligned}(0, c, c) &= c(0, 1, 1) \\ (d, -d, 0) &= d(1, -1, 0)\end{aligned}$$

So:

$$(d, c-d, c) = c(0,1,1) + d(1,-1,0)$$

A spanning set is:

$$\{ (0,1,1), (1,-1,0) \}$$

The two vectors are not scalar multiples, so they are linearly independent.

Conclusion:

$$\dim(W) = 2$$

Part (b):

Start with:

$$(2b, b, 0)$$

Factor:

$$(2b, b, 0) = b(2, 1, 0)$$

A basis is:

$$\{ (2,1,0) \}$$

Since this is one nonzero vector, it is linearly independent.

Conclusion:

$$\dim(W) = 1$$

Example 4.5.10 - Dimension of a subspace of $\mathbb{R}^4$

Let W be spanned by:

$$v_1 = (-1, 2, 5, 0)$$

$$v_2 = (3, 0, 1, -2)$$

$$v_3 = (-5, 4, 9, 2)$$

Find $\dim(W)$.

Solution

Check whether v_3 depends on v_1 and v_2 .

Compute:

$$\begin{aligned} 2v_1 &= 2(-1, 2, 5, 0) \\ &= (-2, 4, 10, 0) \end{aligned}$$

Then:

$$\begin{aligned} 2v_1 - v_2 &= (-2, 4, 10, 0) - (3, 0, 1, -2) \\ &= (-5, 4, 9, 2) \\ &= v_3 \end{aligned}$$

So:

$$v_3 = 2v_1 - v_2$$

Therefore v_3 is unnecessary.

The set:

$$\{v_1, v_2\}$$

spans W .

The two vectors are not scalar multiples, so they are linearly independent.

Final answer:

$$\dim(W) = 2$$

Example 4.5.11 - Dimension of symmetric 2 by 2 matrices

Let W be the subspace of all symmetric 2 by 2 matrices. Find $\dim(W)$.

Solution

A general symmetric 2 by 2 matrix is:

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Separate it:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

So a spanning set is:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Check independence:

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This gives:

$$\begin{bmatrix} c_1 & c_2 \\ c_2 & c_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

So:

$$c_1 = c_2 = c_3 = 0$$

The basis has 3 matrices.

Final answer:

$$\dim(W) = 3$$

Example 4.5.12 - Basis test in $M_{\{5,1\}}$

Show that the following five vectors form a basis for $M_{\{5,1\}}$:

$$\begin{aligned}v_1 &= [1, 2, -1, 3, 4]^T \\v_2 &= [0, 1, 3, -2, 3]^T \\v_3 &= [0, 0, 2, -1, 5]^T \\v_4 &= [0, 0, 0, 2, -3]^T \\v_5 &= [0, 0, 0, 0, -2]^T\end{aligned}$$

Solution

$M_{\{5,1\}}$ has dimension:

$$\dim(M_{\{5,1\}}) = 5 * 1 = 5$$

The set has exactly 5 vectors.

So it is enough to show linear independence.

Solve:

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + c_5v_5 = 0$$

Equating components gives:

$$\begin{aligned}c_1 &= 0 \\2c_1 + c_2 &= 0 \\-c_1 + 3c_2 + 2c_3 &= 0 \\3c_1 - 2c_2 - c_3 + 2c_4 &= 0 \\4c_1 + 3c_2 + 5c_3 - 3c_4 - 2c_5 &= 0\end{aligned}$$

Solve from top to bottom:

$$\begin{aligned}c_1 &= 0 \\c_2 &= 0 \\c_3 &= 0 \\c_4 &= 0 \\c_5 &= 0\end{aligned}$$

Only the trivial solution exists.

Final conclusion:

The set is linearly independent and has 5 vectors in a 5-dimensional space.
Therefore it is a basis for $M_{\{5,1\}}$.

4.6 Rank of a Matrix and Systems of Linear Equations

Example 4.6.1 - Row vectors and column vectors

Let:

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 3 & 4 \end{bmatrix}$$

Find the row vectors and column vectors.

Solution

Rows are read horizontally:

$$\begin{aligned} \text{row 1} &= (0, 1, -1) \\ \text{row 2} &= (-2, 3, 4) \end{aligned}$$

Columns are read vertically:

$$\begin{aligned} \text{column 1} &= [0, -2]^T \\ \text{column 2} &= [1, 3]^T \\ \text{column 3} &= [-1, 4]^T \end{aligned}$$

Final answer:

$$\begin{aligned} \text{Rows:} & \quad (0, 1, -1), (-2, 3, 4) \\ \text{Columns:} & \quad [0, -2]^T, [1, 3]^T, [-1, 4]^T \end{aligned}$$

Example 4.6.2 - Finding a basis for a row space

Find a basis for the row space of:

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{bmatrix}$$

Solution

Row-reduce A to row-echelon form.

The row-echelon form is:

$$B = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The nonzero rows of B form a basis for the row space of A.

Therefore a basis is:

$$\{ (1, 3, 1, 3), (0, 1, 1, 0), (0, 0, 0, 1) \}$$

Final answer:

$$\text{Basis for row space} = \{ (1,3,1,3), (0,1,1,0), (0,0,0,1) \}$$

Example 4.6.3 - Finding a basis for a subspace

Find a basis for the subspace of $\mathbb{R}^3$ spanned by:

$$S = \{ (-1,2,5), (3,0,3), (5,1,8) \}$$

Solution

Put the vectors as rows:

$$A = \begin{bmatrix} -1 & 2 & 5 \\ 3 & 0 & 3 \\ 5 & 1 & 8 \end{bmatrix}$$

Multiply row 1 by -1:

$$\begin{bmatrix} 1 & -2 & -5 \\ 3 & 0 & 3 \\ 5 & 1 & 8 \end{bmatrix}$$

Eliminate below the first pivot:

$$\begin{aligned} R2 &\leftarrow R2 - 3R1 \\ R3 &\leftarrow R3 - 5R1 \end{aligned}$$

This gives:

$$\begin{bmatrix} 1 & -2 & -5 \\ 0 & 6 & 18 \\ 0 & 11 & 33 \end{bmatrix}$$

Make row 2 pivot 1:

$$R2 \leftarrow (1/6)R2$$

$$\begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 11 & 33 \end{bmatrix}$$

Eliminate below row 2:

$$R3 \leftarrow R3 - 11R2$$

$$\begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

The nonzero rows form a basis:

$$\{ (1, -2, -5), (0, 1, 3) \}$$

Example 4.6.4 - Column space basis by Method 1

For the same matrix A from Example 4.6.2, find a basis for the column space by using A^T .

Solution

The column space of A is the row space of A^T .

Row-reducing A^T gives nonzero rows:

$(1, 0, -3, 3, 2)$
 $(0, 1, 9, -5, -6)$
 $(0, 0, 1, -1, -1)$

Interpreting these as column vectors, a basis for the column space is:

$\{ [1, 0, -3, 3, 2]^T, [0, 1, 9, -5, -6]^T, [0, 0, 1, -1, -1]^T \}$

Example 4.6.5 - Column space basis by pivot columns

For the same matrix A , use pivot columns to find a column-space basis.

Solution

The row-echelon form is:

$B = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

The pivot columns are:

1, 2, and 4

Use columns 1, 2, and 4 from the original matrix A :

column 1 = $[1, 0, -3, 3, 2]^T$
column 2 = $[3, 1, 0, 4, 0]^T$
column 4 = $[3, 0, -1, 1, -2]^T$

Final answer:

Basis for column space =
 $\{ [1, 0, -3, 3, 2]^T, [3, 1, 0, 4, 0]^T, [3, 0, -1, 1, -2]^T \}$

Important warning:

Use the pivot columns from the original matrix A , not from the reduced matrix B .

Example 4.6.6 - Finding the rank of a matrix

Find the rank of:

$A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 2 & 1 & 5 & -3 \\ 0 & 1 & 3 & 5 \end{bmatrix}$

Solution

A row-echelon form is:

$$B = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Count the nonzero rows:

3 nonzero rows

So:

$$\text{rank}(A) = 3$$

Example 4.6.7 - Finding the nullspace of a matrix

Find the nullspace of:

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

A row-echelon form is:

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution

Solve:

$$Ax = 0$$

The equations are:

$$\begin{aligned} x_1 + 2x_2 + 3x_4 &= 0 \\ x_3 + x_4 &= 0 \end{aligned}$$

Free variables:

$$\begin{aligned} x_2 &= s \\ x_4 &= t \end{aligned}$$

Solve for leading variables:

$$\begin{aligned} x_1 &= -2s - 3t \\ x_3 &= -t \end{aligned}$$

So:

$$\begin{aligned} x &= [x_1, x_2, x_3, x_4]^T \\ &= [-2s - 3t, s, -t, t]^T \end{aligned}$$

Separate parameters:

$$x = s[-2, 1, 0, 0]^T + t[-3, 0, -1, 1]^T$$

Final answer:

$$N(A) = \text{span}\{ [-2, 1, 0, 0]^T, [-3, 0, -1, 1]^T \}$$

So:

$$\begin{aligned}\text{nullity}(A) &= 2 \\ \text{rank}(A) &= 2\end{aligned}$$

Example 4.6.8 - Rank, nullity, and column-space basis

Let:

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix}$$

and its reduced row-echelon form be:

$$B = \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find rank, nullity, and a basis for the column space.

Solution

Rank:

$$\begin{aligned}B \text{ has 3 nonzero rows} \\ \text{rank}(A) &= 3\end{aligned}$$

Nullity:

A has 5 columns, so:

$$\begin{aligned}\text{number of variables} &= 5 \\ \text{rank} + \text{nullity} &= 5 \\ 3 + \text{nullity} &= 5 \\ \text{nullity} &= 2\end{aligned}$$

Pivot columns in B are:

1, 2, and 4

Use columns 1, 2, and 4 from the original A:

$$\begin{aligned}\text{column 1} &= [1, 0, -2, 0]^T \\ \text{column 2} &= [0, -1, -1, 3]^T \\ \text{column 4} &= [1, 1, -1, 0]^T\end{aligned}$$

Final answer:

$$\begin{aligned}\text{rank}(A) &= 3 \\ \text{nullity}(A) &= 2 \\ \text{basis for column space} &= \\ &\{ [1, 0, -2, 0]^T, [0, -1, -1, 3]^T, [1, 1, -1, 0]^T \}\end{aligned}$$

Example 4.6.9 - Nonhomogeneous system and $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$

Solve:

$$\begin{aligned}x_1 - 2x_3 + x_4 &= 5 \\3x_1 + x_2 - 5x_3 &= 8 \\x_1 + 2x_2 - 5x_4 &= -9\end{aligned}$$

Solution

The augmented matrix row-reduces to:

$$\begin{bmatrix} 1 & 0 & -2 & 1 & | & 5 \\ 0 & 1 & 1 & -3 & | & -7 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

The equations are:

$$\begin{aligned}x_1 - 2x_3 + x_4 &= 5 \\x_2 + x_3 - 3x_4 &= -7\end{aligned}$$

Let the free variables be:

$$\begin{aligned}x_3 &= s \\x_4 &= t\end{aligned}$$

Then:

$$\begin{aligned}x_1 - 2s + t &= 5 \\x_1 &= 5 + 2s - t\end{aligned}$$

and:

$$\begin{aligned}x_2 + s - 3t &= -7 \\x_2 &= -7 - s + 3t\end{aligned}$$

So:

$$\begin{aligned}x &= [x_1, x_2, x_3, x_4]^T \\&= [5 + 2s - t, -7 - s + 3t, s, t]^T\end{aligned}$$

Separate:

$$\begin{aligned}x &= [5, -7, 0, 0]^T \\&\quad + s[2, -1, 1, 0]^T \\&\quad + t[-1, 3, 0, 1]^T\end{aligned}$$

Therefore:

$$\begin{aligned}x_p &= [5, -7, 0, 0]^T \\x_h &= s[2, -1, 1, 0]^T + t[-1, 3, 0, 1]^T \\x &= x_p + x_h\end{aligned}$$

Example 4.6.10 - Consistency using column space

Consider the system:

$$\begin{aligned}x_1 + x_2 - x_3 &= -1 \\x_1 + x_3 &= 3 \\3x_1 + 2x_2 - x_3 &= 1\end{aligned}$$

Show that it is consistent by checking whether b is in the column space of A .

Solution

The columns of A are:

$$\begin{aligned}a_1 &= [1, 1, 3]^T \\ a_2 &= [1, 0, 2]^T \\ a_3 &= [-1, 1, -1]^T\end{aligned}$$

The right-hand side is:

$$b = [-1, 3, 1]^T$$

Check:

$$b = 2a_1 - 2a_2 + a_3$$

Compute:

$$\begin{aligned}2a_1 &= [2, 2, 6]^T \\ -2a_2 &= [-2, 0, -4]^T \\ a_3 &= [-1, 1, -1]^T\end{aligned}$$

Add:

$$\begin{aligned}2a_1 - 2a_2 + a_3 &= [2, 2, 6]^T + [-2, 0, -4]^T + [-1, 1, -1]^T \\ &= [-1, 3, 1]^T \\ &= b\end{aligned}$$

So b is in the column space of A.

Final conclusion:

The system is consistent.

4.7 Coordinates and Change of Basis

Example 4.7.1 - Coordinates relative to the standard basis

Find the coordinate matrix of:

$$x = (-2, 1, 3)$$

relative to the standard basis of $\mathbb{R}^3$.

Solution

The standard basis is:

$$S = \{ (1,0,0), (0,1,0), (0,0,1) \}$$

Write x as:

$$x = -2(1,0,0) + 1(0,1,0) + 3(0,0,1)$$

The coefficients are:

$$-2, 1, 3$$

Final answer:

$$[x]_S = [-2, 1, 3]^T$$

Example 4.7.2 - From a nonstandard basis to the standard basis

The coordinate matrix of x relative to:

$$B = \{ (1,0), (1,2) \}$$

is:

$$[x]_B = [3, 2]^T$$

Find x relative to the standard basis.

Solution

The coordinates mean:

$$x = 3(1,0) + 2(1,2)$$

Compute:

$$3(1,0) = (3,0)$$

$$2(1,2) = (2,4)$$

Add:

$$\begin{aligned} x &= (3,0) + (2,4) \\ &= (5,4) \end{aligned}$$

Relative to the standard basis:

$$[x]_{\text{standard}} = [5, 4]^T$$

Example 4.7.3 - Coordinates relative to a nonstandard basis

Find $[x]_{B'}$ for:

$$x = (1, 2, -1)$$

$$B' = \{ (1, 0, 1), (0, -1, 2), (2, 3, -5) \}$$

Solution

Set:

$$(1, 2, -1) = c_1(1, 0, 1) + c_2(0, -1, 2) + c_3(2, 3, -5)$$

The right side is:

$$(c_1 + 2c_3, -c_2 + 3c_3, c_1 + 2c_2 - 5c_3)$$

Equate components:

$$c_1 + 2c_3 = 1$$

$$-c_2 + 3c_3 = 2$$

$$c_1 + 2c_2 - 5c_3 = -1$$

From the first equation:

$$c_1 = 1 - 2c_3$$

From the second:

$$c_2 = -2 + 3c_3$$

Substitute into the third:

$$(1 - 2c_3) + 2(-2 + 3c_3) - 5c_3 = -1$$

$$1 - 2c_3 - 4 + 6c_3 - 5c_3 = -1$$

$$-3 - c_3 = -1$$

$$-c_3 = 2$$

$$c_3 = -2$$

Then:

$$c_1 = 1 - 2(-2) = 5$$

$$c_2 = -2 + 3(-2) = -8$$

Final answer:

$$[x]_{B'} = [5, -8, -2]^T$$

Check:

$$\begin{aligned} & 5(1, 0, 1) - 8(0, -1, 2) - 2(2, 3, -5) \\ &= (5, 0, 5) + (0, 8, -16) + (-4, -6, 10) \\ &= (1, 2, -1) \end{aligned}$$

Example 4.7.4 - Transition matrix from B to B' in $\mathbb{R}^3$

Find the transition matrix from the standard basis B to:

$$B' = \{ (1, 0, 1), (0, -1, 2), (2, 3, -5) \}$$

Solution

The matrix with the B' vectors as columns is:

$$B' = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \\ 1 & 2 & -5 \end{bmatrix}$$

Since B is the standard basis, the transition matrix from B to B' is:

$$(B')^{-1}$$

Using Gauss-Jordan elimination:

$$[B' \mid I] \rightarrow [I \mid (B')^{-1}]$$

The result is:

$$(B')^{-1} = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -7 & -3 \\ 1 & -2 & -1 \end{bmatrix}$$

So:

$$P^{-1} = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -7 & -3 \\ 1 & -2 & -1 \end{bmatrix}$$

Check with $x = (1, 2, -1)$:

$$P^{-1}[1, 2, -1]^T = [5, -8, -2]^T$$

Example 4.7.5 - Transition matrix in $\mathbb{R}^2$

Find the transition matrix from B to B' for:

$$B = \{ (-3, 2), (4, -2) \}$$

$$B' = \{ (-1, 2), (2, -2) \}$$

Solution

Form:

$$\begin{array}{l} [B' \mid B] \\ = \\ \begin{bmatrix} -1 & 2 & -3 & 4 \\ 2 & -2 & 2 & -2 \end{bmatrix} \end{array}$$

Use Gauss-Jordan elimination to transform:

$$[B' \mid B] \rightarrow [I \mid P^{-1}]$$

The result is:

$$\begin{bmatrix} I & | & P^{-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & | & -1 & -2 \\ 0 & 1 & | & 2 & 3 \end{bmatrix}$$

So the transition matrix from B to B' is:

$$P^{-1} = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$$

Example 4.7.6 - Coordinate representation in P_3

Find the coordinate matrix of:

$$p = 4 - 2x^2 + 3x^3$$

relative to:

$$S = \{1, x, x^2, x^3\}$$

Solution

Write p using every basis vector in order:

$$p = 4(1) + 0(x) + (-2)(x^2) + 3(x^3)$$

The coefficients are:

$$4, 0, -2, 3$$

Final answer:

$$[p]_S = [4, 0, -2, 3]^T$$

Example 4.7.7 - Coordinate representation in $M_{\{3,1\}}$

Find the coordinate matrix of:

$$X = [-1, 4, 3]^T$$

relative to the standard basis for $M_{\{3,1\}}$:

$$S = \{ [1,0,0]^T, [0,1,0]^T, [0,0,1]^T \}$$

Solution

Write X as:

$$X = -1[1,0,0]^T + 4[0,1,0]^T + 3[0,0,1]^T$$

Therefore:

$$[X]_S = [-1, 4, 3]^T$$

4.8 Applications of Vector Spaces

Example 4.8.1 - A second-order linear differential equation

Show that:

$$\begin{aligned}y_1 &= e^x \\ y_2 &= e^{-x}\end{aligned}$$

are solutions of:

$$y'' - y = 0$$

Solution

For:

$$y_1 = e^x$$

Differentiate:

$$\begin{aligned}y_1' &= e^x \\ y_1'' &= e^x\end{aligned}$$

Substitute:

$$y_1'' - y_1 = e^x - e^x = 0$$

So y_1 is a solution.

For:

$$y_2 = e^{-x}$$

Differentiate:

$$\begin{aligned}y_2' &= -e^{-x} \\ y_2'' &= e^{-x}\end{aligned}$$

Substitute:

$$y_2'' - y_2 = e^{-x} - e^{-x} = 0$$

So y_2 is a solution.

Final conclusion:

$$e^x \text{ and } e^{-x} \text{ are both solutions.}$$

Also, any linear combination:

$$y = C_1 e^x + C_2 e^{-x}$$

is a solution because:

$$y'' = C_1 e^x + C_2 e^{-x} = y$$

So:

$$y'' - y = 0$$

Example 4.8.2 - Finding the Wronskian

Find the Wronskian for each set.

- (a) $\{1 - x, 1 + x, 2 - x\}$
(b) $\{x, x^2, x^3\}$

Solution

Part (a):

Let:

$$\begin{array}{lll} y_1 = 1 - x & y_1' = -1 & y_1'' = 0 \\ y_2 = 1 + x & y_2' = 1 & y_2'' = 0 \\ y_3 = 2 - x & y_3' = -1 & y_3'' = 0 \end{array}$$

Wronskian:

$$W = \begin{vmatrix} 1-x & -1 & 0 \\ 1+x & 1 & 0 \\ 2-x & -1 & 0 \end{vmatrix}$$

The third column is all zeros, so:

$$W = 0$$

Part (b):

Let:

$$\begin{array}{lll} y_1 = x & y_1' = 1 & y_1'' = 0 \\ y_2 = x^2 & y_2' = 2x & y_2'' = 2 \\ y_3 = x^3 & y_3' = 3x^2 & y_3'' = 6x \end{array}$$

Wronskian:

$$W = \begin{vmatrix} x & 1 & 0 \\ x^2 & 2x & 2 \\ x^3 & 3x^2 & 6x \end{vmatrix}$$

Expand along the first row:

$$W = x \begin{vmatrix} 2x & 2 \\ 3x^2 & 6x \end{vmatrix} - 1 \begin{vmatrix} x^2 & 2 \\ x^3 & 6x \end{vmatrix} + 0 \begin{vmatrix} x^2 & 2x \\ x^3 & 3x^2 \end{vmatrix}$$

First determinant:

$$\begin{aligned} & (2x)(6x) - 2(3x^2) \\ &= 12x^2 - 6x^2 \\ &= 6x^2 \end{aligned}$$

Second determinant:

$$\begin{aligned} & x^2(6x) - 2(x^3) \\ &= 6x^3 - 2x^3 \\ &= 4x^3 \end{aligned}$$

Therefore:

$$W = x(6x^2) - 4x^3$$

$$\begin{aligned}
 &= 6x^3 - 4x^3 \\
 &= 2x^3
 \end{aligned}$$

Final answers:

- (a) $W = 0$
 (b) $W = 2x^3$

Example 4.8.3 - Testing solutions for linear independence

Determine whether:

$$\{1, \cos x, \sin x\}$$

is a linearly independent set of solutions of:

$$y'''' + y' = 0$$

Solution

First verify solutions.

For $y = 1$:

$$\begin{aligned}
 y' &= 0 \\
 y'' &= 0 \\
 y'''' + y' &= 0
 \end{aligned}$$

For $y = \cos x$:

$$\begin{aligned}
 y' &= -\sin x \\
 y'' &= -\cos x \\
 y'''' &= \sin x
 \end{aligned}$$

Then:

$$y'''' + y' = \sin x - \sin x = 0$$

For $y = \sin x$:

$$\begin{aligned}
 y' &= \cos x \\
 y'' &= -\sin x \\
 y'''' &= -\cos x
 \end{aligned}$$

Then:

$$y'''' + y' = -\cos x + \cos x = 0$$

Now compute the Wronskian:

$$W = \begin{vmatrix} 1 & 0 & 0 \\ \cos x & -\sin x & -\cos x \\ \sin x & \cos x & -\sin x \end{vmatrix}$$

Expand along the first row:

$$W = 1 \begin{vmatrix} -\sin x & -\cos x \\ \cos x & -\sin x \end{vmatrix}$$

Compute:

$$\begin{aligned}
 W &= (-\sin x)(-\sin x) - (-\cos x)(\cos x) \\
 &= \sin^2 x + \cos^2 x \\
 &= 1
 \end{aligned}$$

Since W is not identically zero:

$\{1, \cos x, \sin x\}$ is linearly independent.

General solution:

$$y = C_1 + C_2 \cos x + C_3 \sin x$$

Example 4.8.4 - A linearly dependent solution set

Determine whether:

$$\{e^x, xe^x, (x+1)e^x\}$$

is linearly independent or dependent.

Solution

Let:

$$\begin{aligned}
 y_1 &= e^x \\
 y_2 &= xe^x \\
 y_3 &= (x+1)e^x
 \end{aligned}$$

Expand y_3 :

$$\begin{aligned}
 y_3 &= (x+1)e^x \\
 &= xe^x + e^x \\
 &= y_2 + y_1
 \end{aligned}$$

So one function is a linear combination of the others.

A dependence equation is:

$$y_1 + y_2 - y_3 = 0$$

The coefficients 1, 1, and -1 are not all zero.

Final conclusion:

The set is linearly dependent.

Example 4.8.5 - Identifying conic sections

Part (a)

Identify:

$$x^2 - 2x + 4y - 3 = 0$$

Solution

Group terms:

$$x^2 - 2x + 4y - 3 = 0$$

Move y and constant terms:

$$x^2 - 2x = -4y + 3$$

Complete the square for x:

Half of -2 is -1.

Square it:

$$(-1)^2 = 1$$

Add and subtract 1:

$$x^2 - 2x = (x - 1)^2 - 1$$

Substitute:

$$(x - 1)^2 - 1 = -4y + 3$$

Add 1:

$$(x - 1)^2 = -4y + 4$$

Factor -4:

$$(x - 1)^2 = -4(y - 1)$$

This has form:

$$(x - h)^2 = 4p(y - k)$$

So:

$$h = 1, k = 1, 4p = -4, p = -1$$

Final answer:

Parabola with vertex (1,1), opening downward.
Focus = (1, 0).

Part (b)

Identify:

$$x^2 + 4y^2 + 6x - 8y + 9 = 0$$

Solution

Group terms:

$$x^2 + 6x + 4y^2 - 8y + 9 = 0$$

Factor 4 from y-terms:

$$x^2 + 6x + 4(y^2 - 2y) + 9 = 0$$

Complete the square for x:

$$x^2 + 6x = (x + 3)^2 - 9$$

Complete the square for y:

$$y^2 - 2y = (y - 1)^2 - 1$$

Substitute:

$$(x + 3)^2 - 9 + 4[(y - 1)^2 - 1] + 9 = 0$$

Simplify:

$$(x + 3)^2 + 4(y - 1)^2 - 4 = 0$$

Move 4:

$$(x + 3)^2 + 4(y - 1)^2 = 4$$

Divide by 4:

$$((x + 3)^2)/4 + (y - 1)^2 = 1$$

Final answer:

Ellipse with center $(-3,1)$.
Major axis horizontal.
Vertices: $(-5,1)$ and $(-1,1)$.
Minor-axis endpoints: $(-3,2)$ and $(-3,0)$.

Example 4.8.6 - Transition matrix for rotation in $\mathbb{R}^2$

Find the coordinates of a point (x,y) relative to:

$$B' = \{ (\cos \theta, \sin \theta), (-\sin \theta, \cos \theta) \}$$

Solution

Place B' vectors as columns:

$$B' = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The inverse is:

$$(B')^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

So:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Multiply:

$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

These are the coordinates relative to the rotated basis.

Example 4.8.7 - Rotation of a conic section

Perform a rotation of axes to eliminate the xy -term in:

$$5x^2 - 6xy + 5y^2 + 14\sqrt{2}x - 2\sqrt{2}y + 18 = 0$$

and identify the conic.

Solution

Compare with:

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

So:

$$\begin{aligned}a &= 5 \\b &= -6 \\c &= 5\end{aligned}$$

Use:

$$\cot(2\theta) = (a - c)/b$$

Substitute:

$$\begin{aligned}\cot(2\theta) &= (5 - 5)/(-6) \\&= 0\end{aligned}$$

So:

$$\begin{aligned}2\theta &= \pi/2 \\ \theta &= \pi/4\end{aligned}$$

Then:

$$\begin{aligned}\cos \theta &= 1/\sqrt{2} \\ \sin \theta &= 1/\sqrt{2}\end{aligned}$$

Use the rotation formulas:

$$\begin{aligned}x &= x'\cos \theta - y'\sin \theta \\&= (x' - y')/\sqrt{2}\end{aligned}$$

$$\begin{aligned}y &= x'\sin \theta + y'\cos \theta \\&= (x' + y')/\sqrt{2}\end{aligned}$$

Compute needed products:

$$\begin{aligned}x^2 &= ((x' - y')^2)/2 \\&= ((x')^2 - 2x'y' + (y')^2)/2\end{aligned}$$

$$\begin{aligned}y^2 &= ((x' + y')^2)/2 \\&= ((x')^2 + 2x'y' + (y')^2)/2\end{aligned}$$

$$\begin{aligned}xy &= ((x' - y')(x' + y'))/2 \\&= ((x')^2 - (y')^2)/2\end{aligned}$$

Substitute quadratic terms:

$$\begin{aligned}5x^2 \text{ gives:} \\(5/2)(x')^2 - 5x'y' + (5/2)(y')^2 \\-6xy \text{ gives:}\end{aligned}$$

$$-3(x')^2 + 3(y')^2$$

$5y^2$ gives:

$$(5/2)(x')^2 + 5x'y' + (5/2)(y')^2$$

Combine:

$$(x')^2 \text{ coefficient: } 5/2 - 3 + 5/2 = 2$$

$$x'y' \text{ coefficient: } -5 + 5 = 0$$

$$(y')^2 \text{ coefficient: } 5/2 + 3 + 5/2 = 8$$

So the quadratic part becomes:

$$2(x')^2 + 8(y')^2$$

Substitute linear terms:

$$\begin{aligned} 14\sqrt{2}x &= 14\sqrt{2}(x' - y')/\sqrt{2} \\ &= 14x' - 14y' \end{aligned}$$

$$\begin{aligned} -2\sqrt{2}y &= -2\sqrt{2}(x' + y')/\sqrt{2} \\ &= -2x' - 2y' \end{aligned}$$

Combine linear terms:

$$\begin{aligned} 14x' - 14y' - 2x' - 2y' \\ = 12x' - 16y' \end{aligned}$$

The transformed equation is:

$$2(x')^2 + 8(y')^2 + 12x' - 16y' + 18 = 0$$

Divide by 2:

$$(x')^2 + 4(y')^2 + 6x' - 8y' + 9 = 0$$

Complete the square:

$$(x')^2 + 6x' + 4[(y')^2 - 2y'] + 9 = 0$$

For x' :

$$(x')^2 + 6x' = (x' + 3)^2 - 9$$

For y' :

$$(y')^2 - 2y' = (y' - 1)^2 - 1$$

Substitute:

$$(x' + 3)^2 - 9 + 4[(y' - 1)^2 - 1] + 9 = 0$$

Simplify:

$$(x' + 3)^2 + 4(y' - 1)^2 - 4 = 0$$

Move 4:

$$(x' + 3)^2 + 4(y' - 1)^2 = 4$$

Divide by 4:

$$((x' + 3)^2)/4 + (y' - 1)^2 = 1$$

Final answer:

The rotated conic is an ellipse.
Center in the $x'y'$ plane: $(-3, 1)$.
Semi-major length: 2.
Semi-minor length: 1.

Quick Formula Sheet for Chapter 4 Examples

Vector addition:

$$(u_1, \dots, u_n) + (v_1, \dots, v_n) = (u_1 + v_1, \dots, u_n + v_n)$$

Scalar multiplication:

$$c(u_1, \dots, u_n) = (cu_1, \dots, cu_n)$$

Subspace test:

W is nonempty.
If u and v are in W, then u + v is in W.
If u is in W and c is a scalar, then cu is in W.

Linear independence test:

$c_1v_1 + c_2v_2 + \dots + c_kv_k = 0$
Only trivial solution $\Rightarrow$ independent.
Nontrivial solution $\Rightarrow$ dependent.

Basis:

Basis = spanning set + linearly independent set

Dimension:

$\dim(\mathbb{R}^n) = n$
 $\dim(P_n) = n + 1$
 $\dim(M_{\{m,n\}}) = mn$

Rank-nullity:

$$\text{rank}(A) + \text{nullity}(A) = \text{number of columns of } A$$

Coordinate matrix:

If $x = c_1v_1 + c_2v_2 + \dots + c_nv_n$,
then $[x]_B = [c_1, c_2, \dots, c_n]^T$.

Wronskian:

$W(y_1, \dots, y_n) = \text{determinant of the matrix of functions and derivatives.}$

Rotation of axes:

$x = x'\cos(\theta) - y'\sin(\theta)$
 $y = x'\sin(\theta) + y'\cos(\theta)$

$$\cot(2\theta) = (a - c)/b$$